SAIKARTHICK CHANDRASEKARAN

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PROFILE SUMMARY

CS graduate student at George Mason University (GPA 3.89/4.0) with production software engineering experience at Ford Motor Company and a full-stack agentic AI platform built with LangGraph, MCP, and React. Targeting Software Engineer and Backend Engineer roles.

EDUCATION

Master of Science in Computer Science, George Mason University **08/2025 – Present**
Fairfax, Virginia, USA | GPA: 3.89/4.0

Bachelor of Engineering in Computer Science, Sri Sairam Engineering College **11/2020 – 05/2024**
Chennai, Tamil Nadu, India | CGPA: 8.95/10.00

WORK EXPERIENCE

Junior Engineer, Ford Motor Private Limited **07/2024 – 06/2025**
Chennai, India

- Built a Python-based secret rotation system across GCP projects, managing 200+ secrets and reducing manual engineering effort.
- Enhanced a shutdown automation platform (Go, JavaScript, PostgreSQL) with multi-service concurrency, saving $2,000/week in operational costs.
- Re-architected a Cloud Run shutdown application with zero downtime, integrating automated vulnerability and license compliance scanning (Fossa, SonarQube, Checkmarx).
- Developed Terraform modules to automate GCP resource provisioning, standardizing infrastructure deployments through Infrastructure-as-Code.
- Designed and implemented serverless backend solutions (Cloud Run, Cloud Functions, Eventarc) on GCP, reducing infrastructure costs.

Software Engineer Intern, Ford Motor Private Limited **01/2024 – 07/2024**
Chennai, India

- Evaluated Google Cloud Platform fundamentals and serverless computing principles, resolving the root cause of five application issues.
- Streamlined a proof-of-concept, integrating new features into serverless services to address key customer pain points and increase service adoption by 20%.
- Piloted serverless feature deployment across GCP, reducing deployment time by 15%.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, JavaScript, React, Go, Flask, FastAPI
AI/ML & GenAI: LangGraph, MCP, GenAI Tool Calling, Prompt Engineering, Agentic AI
Databases: MySQL, PostgreSQL, AWS RDS (MySQL), SQLModel, PyMySQL
Cloud & DevOps: GCP (Cloud Run, Functions, Eventarc), AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Terraform, CI/CD
Tools & Practices: Git, Microservices Architecture, Vulnerability Management (Checkmarx, SonarQube), Secret Management

PROJECTS

Agentic Student Survey Management System | LLM, Microservices & React
George Mason University
Production-grade agentic AI system, not a class script — built on a decoupled microservices architecture with a stateful LLM agent, a custom MCP tool backend, and a Kubernetes-orchestrated deployment.

- Built a stateful conversational intake agent using LangGraph and the Google Gemini API, orchestrating multi-turn field collection and structured batch parsing through a defined agent workflow.
- Implemented a secure remote MCP backend with FastAPI, enabling discovery-based tool integration via SSE transport so the agent could dynamically invoke backend capabilities across distinct cluster pods.
- Built a natural-language-to-SQL query engine using SQLModel and PyMySQL, supporting keyword filters and context-preserving updates against an AWS RDS MySQL database, with a strict delete-safety policy in the agent graph that intercepts destructive requests and requires explicit user confirmation before committing changes.
- Built the React frontend for a live interface into the conversational agent and survey administration, and deployed the full system as a multi-tier, decoupled microservices architecture on a single-node RKE2 Kubernetes cluster hosted on AWS EC2.

Leave Management Portal | Full-Stack Web Application

- Designed and developed a web application with a digital approval workflow for college-wide faculty and student leave management, built with HTML, CSS, PHP, and MySQL.

CERTIFICATIONS

RedHat Certified System Administrator (RHCSA) | Google Cloud Certified Associate Cloud Engineer

AWARDS & ACHIEVEMENTS

- 2nd Place, "KLUG KODERS 2.0: Innovative Design Project Exhibition," CSE Department (05/2023).
- Inspirational Star Award for academic excellence (2021–2024), Sri Sairam Engineering College.

PUBLICATIONS

Iswarya M, Saikarthick C, Sudarshan K, "Leave Management Portal," IEEE Xplore – ICCST 2022.